

Response to Reviewer 2 — NLP-D-26-00451

Manuscript: Evaluating LLM-Based Credit Rating Systems: A Pre-Registered Specification-Curve Analysis

We thank the reviewer for the careful reading and for recognizing the “commendably rigorous experimental execution and an exemplary reproducible pipeline.” Each concern is addressed below; every number cited is reproducible offline via `reproduce.sh` (seed 42).

Reviewer comment.

The paper applies a pre-registered specification-curve analysis to evaluate the sensitivity of LLM-generated corporate credit ratings across variations in prompt configurations, formatting choices, and sampling parameters, offering commendably rigorous experimental execution and an exemplary reproducible pipeline. However, the submission suffers from a fundamental domain misalignment for an NLP venue, as its core task—mapping structured financial data to credit ratings—treats the model as a financial classification engine without delivering insights into language representations, semantic mechanisms, or generation dynamics. Furthermore, its empirical claims are constrained by a small underlying item sample ($N = 45$ synthetic profiles and $N = 31$ real-world issuers), severe statistical underpowering in its equivalence testing, and aliasing artifacts in its fractional factorial design. The proposed Specification-Instability Score (SIS) largely rebrands the standard Gini-Simpson dispersion index, offering limited methodological innovation, while comparing basic elicitations against a rigid mathematical model like Altman Z'' without evaluating advanced pipelines like RAG or Chain-of-Thought artificially amplifies variance. Overall, the authors are strongly advised to expand their issuer dataset, evaluate modern reasoning architectures, and re-target the manuscript to a quantitative finance or financial technology venue.

Response. The comment raises seven distinct points; we take each in turn.

1. Fit for an NLP venue / “financial classification engine.”

(Addressed in manuscript Section 1, Section 2.4, and Section 5.)

- The object of study is LLM elicitation reliability: whether an LLM’s categorical judgment remains stable under defensible changes to the linguistic and procedural specification while the substantive input is held fixed.
- We do not claim to identify internal representation mechanisms; our contribution is behavioral evaluation of NLP systems under controlled linguistic and procedural perturbations, analogous to robustness and reliability evaluation in other NLP settings.
- This category of contribution is well within the Journal’s stated scope — the “development and application of trustworthy AI to analyze, process, or model human language across various contexts, domains, and intelligent systems” — and it is

consistent with work the Journal already publishes. Recent examples of behavioral LLM evaluation in the Journal that likewise do not analyze internal representations include Singh & Siami Namin (2025), *A survey on chatbots and large language models: Testing and evaluation techniques*; Espejel et al. (2023), *GPT-3.5, GPT-4, or BARD? Evaluating LLMs’ reasoning ability in a zero-shot setting and performance boosting through prompts*; and de Wynter et al. (2023), *An evaluation on large language model outputs: Discourse and memorization*. Prompt- and paraphrase-sensitivity is also established subject matter here, e.g. Lau & Zubiaga (2025), *Understanding the effects of human-written paraphrases in LLM-generated text detection*, and Al Nazi et al. (2025), *Evaluation of open and closed-source LLMs ... with zero-shot, few-shot, and chain-of-thought prompting*.

- We have revised the introduction and discussion to foreground this as an evaluation-methodology contribution on the specification-sensitivity of LLM outputs, with the instability score offered as a reusable evaluation tool. The credit-rating setting is the testbed, not the contribution.

2. Small item sample (N=45 synthetic, N=31 real).

(Addressed in manuscript Section 5.2; clustering in the Table 1–5 captions and Methods Section 6.3.)

- We agree that the number of independent underlying credit profiles is 45 in the controlled arm and 31 in the real-issuer arm. The design is intended to estimate within-profile specification sensitivity, not population prevalence. The repeated elicitation conditions provide dense paired measurement of that within-profile estimand, while uncertainty is clustered at the profile/issuer level.
- We have revised the manuscript to make this distinction explicit and to avoid treating the 8,640 elicitation cells as independent observations; confidence intervals are profile-clustered on the constructed battery (issuer-clustered on the real-issuer arm).
- We have also added a severity analysis that uses the profile structure informatively (stratum-level and boundary-distance breakdowns).

3. Underpowering in the equivalence testing.

(Addressed in manuscript Section 3.6, Table 4, and Methods Section 6.3.)

- We agree. The equivalence (TOST) analysis has been **removed**. The real-firm arm is now presented purely descriptively, with explicit sample-size caveats and no equivalence claim.

4. Aliasing in the fractional factorial design.

(Addressed in manuscript Section 3.1, Table 1, Table 2, and Section 5.2.)

- We now report the aliasing diagnostics from the frozen design directly (computed offline in `analysis/crossmodel_aliasing.py`): main-effect columns are near-orthogonal (maximum off-diagonal correlation **0.062**), while the maximum absolute correlation between a main effect and a two-factor-interaction of other factors is **0.333**, confirming that non-negligible aliasing remains. For this reason the decomposition is retained only descriptively.

- The variance components are re-presented as descriptive partitions under the assumed model, not causal attributions, and per-axis figures are qualified accordingly.

5. The instability score “rebrands the Gini-Simpson index.”

(Addressed in manuscript Section 4, Section 4.2, and Table 7.)

- We agree the underlying quantity is the Gini-Simpson concentration ($1 - \sum p_c^2$), and we now credit this explicitly.
- The contribution is not a new index but its operationalization and validation as a candidate operational, ground-truth-free control: a held-out-specification validation framework (ROC-AUC **0.94** on the constructed battery; median **0.88** across 300 splits; **0.70** on real issuers), a practical descriptive operating point (a k -dependent operating rule), and an honest account of cross-model transfer, which is moderate but inconsistent across model pairs (per-item Spearman 0.14-0.73, median ≈ 0.45 ; AUC ≈ 0.72), too variable to rely on, for which we state the concrete recalibration requirement — a new model should be re-scored on the specification grid (about **\$7.43 per 1,000 issuers at $k = 9$ on a nano/flash tier**, from the deployment cost table) rather than reusing another model’s scores. Section 4 has been rewritten to make this scope precise.

6. No advanced pipelines (RAG / Chain-of-Thought); comparison to a rigid Altman Z” amplifies variance.

(Addressed in manuscript Section 3.8, Table 6, Table 9, and Section 5.3.)

- We have added a reasoning-architecture arm using an open-weight model with native chain-of-thought (DeepSeek-R1), on the 45-item $\times$ 3-seed grid across the frozen 12 specifications. We are precise about what this arm varies. The frozen 12-specification grid spans four axes — provider, prompt paraphrase, output format, and answer presentation (version, few-shot, and temperature are fixed). A single model instantiates the grid, so the reasoning arm exercises the three non-provider axes — paraphrase (3 levels), output format (2), and presentation (2), fully crossed into all 12 distinct configurations — while holding the fourth axis, provider, constant. It is therefore not literally the two-provider grid, and we describe it accurately: the +0.9-percentage-point paired difference below compares three axes of variation (reasoning) against four (nano/flash). The arm is labelled post-hoc (not pre-registered); it was collected via the OpenRouter API (model deepseek/deepseek-r1) and frozen with a SHA-256 manifest, and reported with a dedicated results table, a new row in the model-snapshots table, a manifest entry, and a line in the data-availability statement.
- Result: the reasoning-model arm shows a **25.5% IG/HY flip rate** (95% CI 20.9–30.1%). The paired difference against the nano/flash tier is **+0.9 percentage points** (95% CI **−7.0 to +8.8**), i.e. statistically indistinguishable, and comparable to the flagship tier (23.6%). We therefore do not observe evidence of attenuation under explicit reasoning in this arm.
- Two design differences qualify the comparison. First, the reasoning arm contains no cross-provider variation, which §3.3 shows is the largest agreement gap (band-level agreement 64.4% across providers versus 82.5% within); the nano/flash 24.7% and flagship 23.6% figures include that component, whereas

DeepSeek’s 25.5% does not. Second, DeepSeek-R1 has a substantially more skewed marginal IG/HY distribution (31.4% investment-grade versus 79.9% for the flagship tier). Both features change the attainable disagreement structure. We therefore interpret the 25.5% flip rate as evidence of specification sensitivity within the evaluated reasoning model, rather than as a strictly like-for-like comparison across tiers.

- On amplification: Altman Z” is used only as a pre-specified construction and stratification reference. The core instability measures (IG/HY flip, cross-specification agreement) are computed from within-model disagreement and use no benchmark, so they cannot be inflated by the choice of benchmark.
- On RAG specifically: retrieval augmentation changes the information set available to the model rather than the elicitation of a fixed input, so it would break the controlled design that holds the substantive facts constant across specifications. We therefore treat retrieval-augmented pipelines as out of scope for this controlled study and note them as future work; the chain-of-thought arm is the pipeline variation compatible with the design.

7. Suggested re-targeting to a finance venue.

(Addressed in manuscript Section 1 and Section 2.4.)

- With the strengthened elicitation-reliability framing (point 1), the reasoning-architecture arm (point 6), and the reusable evaluation score (point 5), the contribution sits as an LLM evaluation-methodology paper, and we have kept it here.
- We would also note, respectfully, that a finance *application domain* does not place the work outside the Journal’s scope. The Journal publishes finance-domain NLP — e.g. Sun et al. (2025), *Financial sentiment analysis for pre-trained language models incorporating dictionary knowledge and neutral features*, and Chen et al. (2025), *Sentiment analysis for stock market research: A bibliometric study* — and, separately, LLM reliability and evaluation studies, including domain-situated LLM evaluations in e-commerce, radiology, and clinical records. To our knowledge, this is the first study in the Journal to evaluate elicitation-specification sensitivity in direct LLM-generated credit-rating judgments, which we see as complementing these established strands rather than falling outside the Journal’s scope.
- Moreover, the underlying evaluation framework is not intrinsically finance-specific and can in principle be applied to other categorical LLM judgment tasks, so the contribution is not narrowly domain-bound. We are of course glad to defer to the editor on venue fit.

Response to Reviewer 3 — NLP-D-26-00451

Manuscript: Evaluating LLM-Based Credit Rating Systems: A Pre-Registered Specification-Curve Analysis

We thank the reviewer for the thorough reading and for the accurate summary of the contribution. We have adopted the recommended option for each required revision. Every number cited is reproducible offline via `reproduce.sh` (seed 42).

Required revision 1 — Circularity between constructed profiles and the Altman benchmark

Reviewer comment.

the authors design their 45 credit-health profiles to correspond to pre-specified altman z' strata, but then use the same z' mapping to define “benchmark-consistent credit-band accuracy” in the specification curve (figure 1). this creates a circularity: the profiles were constructed to align with z' categories, and the models are evaluated against their alignment with those same constructed categories. the specification curve showing that some specifications yield “majority-correct” or “majority-incorrect” credit-band verdicts is therefore not a meaningful measure of model capability or specification sensitivity->it is largely a reflection of how well the models reconstruct the design intent embedded in the profiles themselves. the authors treat this as a demonstration of instability, but it could equally be interpreted as demonstrating that the models sometimes deviate from the constructed labels, which is not the same as showing that they produce unstable assessments of real-world credit risk. the distinction matters because the constructed profiles are not independent of the benchmark->they were generated to satisfy the benchmark strata. the 56/44 split around the majority-accuracy threshold is therefore not a neutral diagnostic.

required revision-> the authors must either (a) construct the credit-health profiles independently of any benchmark and use a separate, truly external reference (e.g., historical agency ratings for real firms) for the accuracy-oriented analyses, or (b) clearly re-frame the specification curve analysis as a diagnostic of how consistently the models recover the design labels rather than as evidence about credit-rating accuracy or instability. if option (b) is chosen, the language throughout (e.g., “benchmark-referenced,” “majority-correct,” “accuracy”) must be revised to avoid implying external validity. additionally, the authors should provide a version of the specification curve computed on a benchmark that is not used in the construction of the profiles, even if this requires a separate real-firm validation sample.

(Addressed in manuscript Section 3.1 and Figure 1; external-benchmark curve in Section 3.6 and Table 4.)

Response. We adopt **option (b)** and reframe the specification-curve analysis as a diagnostic of label recovery.

- The curve is now described as *how consistently models recover the constructed design labels*, not as credit-rating accuracy. Language such as “benchmark-referenced,” “majority-correct,” and “accuracy” has been revised throughout to avoid implying external validity; the 56/44 split is presented as label-recovery consistency.
- Our central instability finding does not use the benchmark at all: the IG/HY flip (**25.2%**) and cross-specification agreement are computed from within-model disagreement across elicitations, with no external label.
- As specifically requested, we now provide a specification curve computed on a benchmark that is **not** used in constructing the profiles: the 31 perturbed and fingerprint-screened real issuers scored against their **disclosed agency ratings** (an external reference, independent of profile construction). Per-specification exact credit-band accuracy against the agency ratings (on the SAFE and WATCH strata, i.e. 30 of the 31 issuers; the single DISTRESS issuer is a singleton excluded from this three-band comparison) ranges from **42.2% to 61.1% across the 12 specifications** (mean 53.7%).
- To test whether the between-specification spread exceeds sampling noise, we use an issuer-preserving permutation test: specification labels are permuted **within each issuer**, preserving the repeated-measures structure, with the statistic being the between-specification standard deviation of per-specification accuracy. The observed spread is beyond chance (observed SD 0.050 vs null mean 0.035, both accuracy proportions; $p = 0.021$). The external-benchmark curve therefore shows specification sensitivity distinguishable from sampling noise, not only against the constructed labels.
- We keep the interpretation measured given the small sample (31 issuers): the confidence interval on the *mean* accuracy is wide, so we lean on the spread test rather than the mean, and the primary evidence for instability remains the benchmark-free within-model IG/HY flip (25.2%), which uses no external label. (Reproducible via `analysis/realarm_speccurve.py`.)

Required revision 2 — Variance components are not identified under the design

Reviewer comment.

the authors use a “balance-constrained d-optimal fractional grid” generated from an effect-coded main-effects model, but they acknowledge that “with n not divisible by the strength-2 requirement the design is resolution-limited: main effects can be partially aliased with two-factor interactions.” this is a serious methodological limitation that the authors mention in the limitations section but do not adequately address in the presentation of results. the decomposition reported in table 1 and discussed in section 3.1 (“no individual controllable elicitation choice explains more than about 5% of rating variance”) is presented with precise-looking confidence intervals, but these estimates are conditional on a design that cannot cleanly separate main effects from interactions. the “residual” of ~25% is therefore not a mean-

ingful residual in the standard sense->it could include substantial main-effect variance that is aliased with omitted interactions. the authors' subsequent claim that "most residual variation is structured sensitivity" is not supported by the design because the design cannot identify whether that structure is due to main effects, two-factor interactions, or higher-order terms.

Required revision: the authors should either (a) collect data on a full factorial or a resolution- v sufficient fractional design that can reliably estimate all main effects and relevant two-factor interactions, or (b) revise the variance decomposition section to clearly state that the component estimates are not identified and should be interpreted only as descriptive partitions under the assumed model, not as causal attributions. if option (b) is chosen, the authors must remove or substantially qualify statements that imply specific design axes have particular effect sizes (e.g., "answer presentation contributes $\sim 4.6\%$ ") and instead present the range of variance explained by different model specifications. the confidence intervals should also be re-computed using a method that accounts for the design's non-orthogonality (e.g., bayesian hierarchical modeling with shrinkage).

(Addressed in manuscript Section 3.1, Table 1, Table 2, and Section 5.2.)

Response. We adopt **option (b)** and implement it thoroughly: the variance decomposition is now clearly secondary, and the paper's conclusions rest on the model-free flip and agreement results.

- We have **removed the inferential confidence intervals from the individual per-axis variance shares** in Table 1. The individual shares are presented only as descriptive partitions under the assumed main-effects model, explicitly not as identified effect sizes or causal attributions, and per-axis magnitudes (e.g., answer presentation $\approx 4.6\%$) are reported without the precise-looking intervals the reviewer rightly flagged.
- We report the aliasing diagnostics from the frozen design (computed offline in `analysis/crossmodel_aliasing.py`): main-effect columns are near-orthogonal (maximum off-diagonal correlation **0.062**); the maximum absolute correlation between a main effect and a two-factor-interaction of other factors is **0.333**, confirming that non-negligible aliasing remains. For this reason the decomposition is retained only descriptively.
- The $\sim 25\%$ residual is described as a descriptive residual that may contain aliased interaction variance; we no longer describe it as identified "structured sensitivity."
- As explicitly requested, we also report the range of variance the elicitation axes explain under different model specifications (new table in §3.1 of the revised manuscript): a main-effects-only model gives $R^2 = \mathbf{0.083}$; adding all two-way interactions raises the in-sample R^2 to **0.119**; an issuer-out-of-sample regularized (ridge) model with the same interactions recovers $R^2 = \mathbf{0.101}$. The apparent explained variance thus shifts with the modelling choice, and all specifications remain small — no modelling choice makes the axes explain much of the rating. This shifting decomposition is itself the reason individual-axis attribution is not identified, and we present the components descriptively only. (Reproducible via

analysis/variance_model_range.py.)

- None of the paper’s substantive conclusions depends on the ANOVA. The headline results — the IG/HY flip (25.2%), cross-specification agreement, the permutation test (Required revision 3), and the specification-instability score (Required revision 5) — are model-free quantities that make no orthogonality assumption. Section 3.1 now states this explicitly and positions the decomposition as a descriptive summary only.
- On the reviewer’s specific suggestion of a Bayesian hierarchical model with shrinkage: we did not fit one, and we now explain why. A shrinkage model would produce refined *inferential* estimates of the per-axis components, but under option (b) we no longer make an inferential or causal decomposition — the components are presented only as descriptive partitions and carry no CIs. Fitting a hierarchical model to obtain more precise estimates of quantities we deliberately no longer interpret inferentially would misrepresent the role of the decomposition in the revised paper. We therefore report the design’s aliasing directly and rest the conclusions on the model-free quantities instead.

Required revision 3 — Permutation test used the wrong null

Reviewer comment.

The authors test specification invariance under “deterministic execution” using the Google temperature=0 setting, computing the observed between-specification spread in credit-band accuracy and comparing it to a permutation distribution. however, the permutation test as described (label-permutation) permutes the labels within each specification, which tests whether the observed spread could arise from random assignment of accuracy outcomes->but this is not the relevant null hypothesis. the relevant null is that ratings are invariant to the specification, which would imply that all specifications produce the same rating for each profile. the permutation test should permute the specification assignments across profiles, not the accuracy labels. the current test is essentially a test of whether accuracy differs across specifications, which is a much weaker condition than specification invariance. moreover, the $p=0.001$ result is reported with only 999 permutations, which is a small number for a one-sided test given the observed effect size->the authors should report the exact number of permuted statistics as extreme as the observed.

required revision: the authors must re-specify and re-run the permutation test with the correct null hypothesis (invariance of ratings across specifications, not invariance of accuracy). the test should permute the specification-condition assignments across profiles and compare the observed between-specification spread in rating agreement to the null distribution. additionally, the authors should provide the full permutation distribution and report the effect size (e.g., the standard deviation of the accuracy spread under permutation) alongside the p-value. if the data from Google temperature=0 are limited (the authors note that “temperature of 0 was not uniformly honored by providers”), the test should be re-run on a subset of the data where the authors can independently verify that temperature=0

was honored across all calls....

(Addressed in manuscript Section 3.2, Table 3, and Methods Section 6.3.)

Response. Corrected as requested.

- The test is re-specified to the rating-invariance null — permuting specification-condition assignments across profiles, not accuracy labels — and re-run with 2000 permutations. The statistic is the between-specification spread (standard deviation) of the item-centred letter rating, so it measures whether ratings depend on the specification, not merely whether accuracy differs.
- Result: observed spread = **1.59 notches**; permutation null mean **0.18** notches, standard deviation **0.050 notches**; **0 of 2000** permuted statistics reach the observed value; **p = 0.0005** (the minimum attainable, reported as the exact exceedance count). The specification η^2 on the item-centred rating is **0.46**. (This 0.050 is in notches; the numerically identical 0.050 in Required revision 1 is an accuracy-proportion standard deviation — an unrelated quantity.)
- We run this test on the temperature-0-honored subgrid (Google, requested temperature 0), which is precisely the subset in which temperature 0 was honoured across all calls, satisfying the reviewer’s closing condition.
- The η^2 and the spread statistic are two summaries of the *same* object: both are computed from the item-centred letter rating on this subgrid. The permutation statistic is the standard deviation, across specifications, of the per-specification mean item-centred rating; the η^2 is the share of the total item-centred-rating variance attributable to the specification factor. We report both so the effect is legible as a spread (1.59 notches) and as a variance share (0.46).
- This η^2 is a specification-level aggregate on the temperature-0-honored subgrid and is therefore a different quantity from the individual-rating variance partition in Table 1 (where issuer identity dominates and no single elicitation axis exceeds ~5%); the two are not directly comparable.

Required revision 4 — Decontamination procedure and the equivalence test

Reviewer comment.

the real-firm decontamination procedure is ad hoc, insufficiently validated, and the quantitative equivalence test is mis-specified... the authors’ real-firm arm uses a “decontamination” procedure that multiplies all dollar-denominated quantities by an issuer-specific factor sampled from log-uniform[0.6, 1.7], arguing that this “disrupts exact-figure lookup” while preserving financial-ratio structure. however, this transformation is not mathematically neutral for credit-risk assessment->it changes the scale of all monetary variables, which affects ratios involving both monetary and non-monetary quantities (e.g., equity/total assets). the transformation also introduces random variation that may itself affect rating stability, and the authors do not demonstrate that the transformed profiles preserve the original credit-risk ordering or that the factor sampling does not systematically bias the results. the fingerprinting test shows that identification falls to 2.9%, but this is a test of exact-figure recognition,

not of broader inference->a model might still infer issuer identity from the pattern of ratios even if it cannot recognize the exact numbers. the tost equivalence test comparing real and constructed watch-stratum flip shares is pre-registered with a margin of ± 0.15 , but the test is underpowered (n=16 real and 15 constructed observations) and the authors acknowledge that “the watch-stratum comparison lacks the power to detect the pre-specified equivalence margin.” reporting a failed equivalence test as a “robustness” check is misleading->failure to reject equivalence is not evidence of equivalence.

required revision: the authors should either (a) develop and validate a more principled decontamination procedure that preserves credit-risk structure while preventing identification, including a demonstration that the transformed profiles preserve the same ranking on credit-risk metrics, or (b) abandon the quantitative equivalence claim and present the real-firm results purely descriptively, with appropriate cautionary language about the sample size and identification limitations. if option (b) is chosen, the authors must remove the tost analysis and the claim that the real-firm results “qualitatively persist”->the appropriate conclusion is that the real-firm sample is too small and too contaminated to provide meaningful external validation. the authors should also conduct a sensitivity analysis varying the perturbation scale to show that the results are not driven by the specific [0.6, 1.7] range.

(Addressed in manuscript Section 3.6, Table 4, and Methods Section 6.4.)

Response. We adopt **option (b)** and add the requested demonstrations.

- The TOST analysis and any “qualitatively persist / equivalent” language are **removed**; the real-firm arm is now descriptive with sample-size and identification caveats.
- The transformation multiplies all dollar quantities by a single per-issuer factor — a uniform rescaling — so every financial ratio, including equity/total assets, is invariant, and therefore Altman Z” is invariant by construction: Z” transformed-vs-original correlation = **1.000000**, and **100%** of issuers remain in their original SAFE/WATCH/DISTRESS stratum.
- We agree, however, that ratio-invariance of Z” does not by itself guarantee invariance of the *LLM*’s output, since the model also sees the absolute dollar figures. We therefore added a perturbation-scale sensitivity analysis. Because each issuer received an independent factor, the realized factors already span the full pre-registered range (0.60–1.66 across the 31 issuers), so the collected data directly vary the monetary scale across [0.6, 1.7]. Testing whether per-issuer instability and rating level track the applied scale, both associations are negligible: instability vs. scale, Spearman **−0.035** (permutation p = **0.85**); rating level vs. scale, Spearman **−0.019** for mean letter-rank and **+0.019** for the investment-grade rate. We do not observe an association between realized monetary scale and either instability or rating level over the tested range.
- We are explicit about two limits of this check. First, it covers only the pre-registered support [0.6, 1.7]; behaviour under wider perturbations is untested, and we did not collect data outside that range. Second, and more important,

each issuer received a single scale draw, so the applied scale is confounded with issuer identity: the Spearman above is a between-issuer correlation and cannot, on its own, separate “scale does not matter” from “issuer-to-issuer differences swamp any scale effect.” A clean test would re-elicited a handful of issuers at two or three different factors (a within-issuer design); we flag this as the appropriate next step and state the confound plainly rather than resting the claim on the between-issuer correlation alone. (Reproducible via `analysis/realarm_scale_sensitivity.py`.)

- The fingerprinting claim is softened to “exact-figure lookup,” acknowledging it does not rule out inference from ratio patterns.

Required revision 5 — Out-of-sample validation of the instability score

Reviewer comment.

the authors propose a specification-instability score (sis) defined as $1 - \sum(p_c^2)$ over the distribution of ratings across specifications, and they validate this score by showing that it predicts held-out instability with roc-auc 0.94 on the constructed battery and 0.70 on real issuers. however, the sis is computed using the same specification grid and the same model calls that are used to evaluate its predictive performance->there is no separate training and test set for the construction of the score itself. the authors state that the score is “lightweight” and “ground-truth-free,” but they do not provide evidence that the score generalizes to new specifications (e.g., prompt paraphrases not in the original grid) or to new models not used in the construction. the roc-auc of 0.94 on the constructed battery is essentially a measure of how well the score captures the variance in the same data from which it was derived, which is not informative about its utility as a deployable control. the real-issuer roc-auc of 0.70 is more informative but is based on a small sample (31 issuers) and is not accompanied by confidence intervals or a cross-validation analysis.

required revision: the authors must re-evaluate the sis using a proper out-of-sample validation framework. this should include (a) a cross-validation analysis on the constructed battery, where the sis is computed on a subset of specifications and evaluated on the held-out specifications; (b) a separate validation on specifications or models not used in the main experiment (e.g., the flagship model tier could be used for this purpose rather than as a robustness check); and (c) a demonstration that the score’s performance is stable across different choices of the specification set (e.g., random subsamples of the 32 specifications). the authors should also provide a practical calibration of the score->what threshold τ should practitioners use, and how does the score’s performance vary with τ ? the current algorithm 1 includes a flag based on a threshold but the threshold is not specified or validated.

(Addressed in manuscript Section 4.2, Algorithm 1 (Figure 5), and Table 7.)

Response. Rebuilt as a proper out-of-sample framework.

- **Held-out-specification cross-validation:** the score is computed on 16 calibration specifications (the SIS has no fitted parameters) and evaluated on the disjoint 16 validation specifications — ROC-AUC **0.94** (profile-clustered bootstrap 95% CI **0.86-0.99**), average precision 0.98, base rate 0.76.
- **Split stability:** median AUC **0.88** across 300 balanced random splits (IQR 0.86–0.91).
- **External validation:** here “external” means external *issuers*, not external specifications — held-out-specification instability on the 31 real issuers (a 6/6 split of the 12-specification grid), ROC-AUC **0.70** (issuer-clustered bootstrap 95% CI **0.51-0.87**, base rate 0.61). We report this interval explicitly; it is wide, and we describe the external-issuer result as suggestive rather than definitive.
- **Cross-model transfer, reported honestly:** moderate but inconsistent across model pairs (per-item Spearman 0.14–0.73, median ≈ 0.45 , and ≈ 0.55 across provider families; AUC of one model’s score flagging another’s above-median-instability items, median ≈ 0.72), computed offline in `analysis/crossmodel_aliasing.py`. Because the transfer is too variable to rely on, the score is treated as per-model/per-grid and we give the concrete recalibration cost: a new model should be re-scored on the specification grid (about **\$7.43 per 1,000 issuers at $k = 9$ on a nano/flash tier**, from the deployment cost table) rather than reusing another model’s scores.
- **Practical operating threshold, stated honestly about discreteness:** at $k = 16$ elicitation the attainable SIS values are discrete — for a binary IG/HY outcome the score jumps from 0 (unanimous) to 0.117 (a single dissenter), with nothing in between. The Youden-optimal threshold on the held-out split, $\tau = \mathbf{0.08}$, therefore lies below 0.117 and simply selects the “flag any issuer with ≥ 1 disagreeing calibration elicitation” rule; it is the floor, not a finely tuned value. Under that rule, held-out precision is **1.00** and recall **0.88**. Because the threshold was chosen and evaluated on the same held-out split, we present it as a descriptive operating point rather than an independently tuned one, and note it is k -dependent (at the recommended $k \approx 9$ the smallest non-zero SIS is 0.198), so practitioners should apply the “flag any disagreement” rule at their chosen k rather than transfer the numeric τ .
- **What that means for referral volume:** the SIS-only rule fires on $\approx 67\%$ of issuers (a direct consequence of the finding that most issuers disagree on at least one specification). This is why Algorithm 1’s flag is a *conjunction* — high SIS **and** a near-boundary modal rating — which restricts referrals to the decision-relevant subset and roughly halves the volume to $\approx 33\%$ of issuers on the battery. We report the post-conjunction rate, since that is the number a practitioner acts on.
- **On the score reducing to a unanimity indicator:** at the binary IG/HY gate with a low τ the flag does reduce to “any disagreement,” but this is a property of the binary trigger, not of the score. The continuous SIS is reported alongside the modal rating for *ranking and triage* (its discrimination — ROC-AUC 0.94 — is computed on the full continuous score, not the binary gate), and at finer resolutions than IG/HY the score takes many values. The Gini-Simpson form is standard; the contribution is its validated use as a candidate operational control in the evaluated model/specification setting, not the index itself.

Required revision 6 — Economic translations

Reviewer comment.

the economic translations (portfolio turnover, spread range, pillar-1 capital) are post-hoc, use arbitrary assumptions, and the capital illustration is actively misleading. the authors describe the economic translations as “post-hoc calibrations” and “not confirmatory tests,” but they are presented in the results section (3.5) alongside confirmatory findings and are given substantial weight in the abstract and introduction. the portfolio turnover calculation assumes a strategy that “mechanically follows the modal buy/hold/sell call across specifications,” which is not a strategy that anyone would implement->it is a theoretical construct that does not correspond to any real decision-making process. the implied spread range of 275 bps is based on the rating-spread schedule of cornaggia et al. (2018), which is a study of municipal bonds, not corporate credit->the applicability to the authors’ corporate credit profiles is unclear. the pillar-1 capital illustration is the most problematic: the authors map ratings through the basel cre20 risk-weight schedule and apply it to a hypothetical \$45 billion equal-weight corporate portfolio, but the basel standardized approach is intended for banks’ regulatory capital calculations, not for credit-rating evaluation, and the mapping from credit ratings to risk weights depends on the specific portfolio characteristics (e.g., seniority, collateral, maturity) that are not specified. the authors’ claim that “two randomly drawn specifications differ by about \$0.57 billion on average” is presented without accounting for the uncertainty in the risk-weight mapping or the arbitrary choice of portfolio size and composition. the comparison to the basel rcap exercise is particularly inappropriate->rcap examined variation in risk-weighted assets across banks with identical portfolios, which is a very different setting from variation in a hypothetical llm-generated rating.

required revision: the authors must either (a) remove the economic translations entirely, as they are not supported by the study design and introduce more confusion than insight, or (b) relegate them to a separate supplementary section with explicit caveats and a clear statement that they are illustrative only and not based on any institutional implementation. if option (b) is chosen, the authors must: (i) justify the choice of the cornaggia et al. (2018) spread schedule for the corporate credit context, or use a corporate-bond spread schedule; (ii) specify the portfolio assumptions (seniority, collateral, maturity) used in the basel mapping and conduct a sensitivity analysis over reasonable variations in these assumptions; and (iii) remove the basel rcap comparison, as it is not comparable. the portfolio turnover calculation should be replaced with a more realistic measure, such as the percentage of cases where the modal call changes when using a different specification.

(Addressed in manuscript Section 3.5, Table 3, Methods Section 6.3, and Section 6.5.)

Response. We agree with the reviewer’s assessment and adopt **option (a)**. All spread, turnover, and regulatory-capital translations — including the Cornaggia-

anchored spread range, the portfolio-turnover construct, the Basel Pillar-1 capital illustration, and the RCAP comparison — have been **removed** from the manuscript. The revised paper now focuses on the empirically identified specification-instability quantities, which we regard as its core contribution.

Required revision 7 — Over-general framing; flagship “no attenuation”

Reviewer comment.

the authors repeatedly frame the study as establishing a general problem with llm-based credit rating instability, but the results are specific to the particular models (gpt-5.4-nano, gpt-5-nano-2025-08-07, gemini-3.5-flash, gemini-2.5-flash) and the particular specification grid used. the authors acknowledge this in the limitations section (“the results should therefore be viewed as evidence of how the evaluated systems behave, not as a universal numerical estimate”), but this caveat is buried in the final pages and is inconsistent with the abstract’s claim that “substantial instability persists on 31 decontaminated real issuers, and a post-hoc check finds no evidence that it attenuates on flagship models”->the flagship check only covers two models (gpt-5.4 and gemini-3.1-pro-preview) on a reduced 12-specification grid and the confidence intervals for the difference are wide (ig/hy flip difference -1.1%, 95% ci -7.6 to +6.0). this is a classic “absence of evidence is not evidence of absence” problem. the authors also do not test whether specification sensitivity varies with the model’s confidence calibration, with the complexity of the financial profile, or with the distance from rating boundaries->all of which would be practically important for understanding when the instability is most consequential.

required revision: the authors should fundamentally re-frame the paper as a case study of specification sensitivity in a particular set of llm systems, with the sis as a method for characterizing that sensitivity, rather than as a general finding about llm credit rating instability. the abstract, introduction, and conclusion should be revised to emphasize the conditional nature of the findings. the authors should also conduct additional analyses to characterize when the instability is most severe->for example, stratifying by altman z’ stratum (safe/watch/distress) or by distance from the ig/hy boundary->and report whether the sis predicts this variation. if the authors want to make a claim about the general problem, they need to test a broader range of models (including frontier models from multiple providers), specifications (more varied paraphrases, different instructions), and profiles (more diverse financial structures) to establish the robustness of the pattern.

(Addressed in manuscript Sections 3.4, 3.7, 3.8, Table 6, Table 9, Section 5.2, and the Conclusion.)

Response. Reframed as a case study.

- The abstract, introduction, and conclusion now emphasize the conditional nature of the findings — specific to the evaluated systems and grid — with the instability score as a method for characterizing that sensitivity.

- The flagship statement is softened from “no evidence it attenuates” to “does not detect attenuation,” explicitly noting the wide interval (IG/HY flip difference -1.1% , 95% CI -7.6 to $+6.0$) and the two-model / 12-specification scope.
- Severity characterization added:
 - By Z'' stratum: IG/HY flip SAFE **13.0%**, WATCH **35.5%**, DISTRESS **27.1%** — instability concentrates mid-scale (WATCH).
 - By distance from the IG/HY boundary: Spearman(flip, $|Z'' - 2.60|$) = **-0.05** — we do not detect a monotonic association between instability and distance from the boundary (reported as a null result).
 - The SIS also predicts issuer-level variation in instability: the calibration-set score is strongly associated with the held-out flip severity (Spearman ρ = **0.89**; ROC-AUC **0.94**), i.e. the score identifies which issuers will be most unstable (§4.2).
- A reasoning-architecture arm (DeepSeek-R1, open-weight, native chain-of-thought) is added, testing robustness across model class. The frozen grid spans four axes (provider, paraphrase, output format, presentation; version/few-shot/temperature fixed); a single model instantiates it, so the arm exercises the three non-provider axes (fully crossed into all 12 configurations) and holds provider constant — we describe it accordingly rather than as the two-provider grid, and the $+0.9\text{pp}$ difference therefore compares three axes against the two-provider arms’ four. Specification sensitivity is still observed (IG/HY flip **25.5%**, 95% CI 20.9–30.1%; paired difference vs nano/flash **+0.9pp**, 95% CI **-7.0 to +8.8**); the comparison does not establish that reasoning has no effect, but it provides no evidence of attenuation in the evaluated model. Two design differences qualify the comparison: the reasoning arm contains no cross-provider variation (the largest agreement gap, §3.3) and DeepSeek-R1 has a substantially more skewed marginal IG/HY distribution (31.4% investment-grade), both of which change the attainable disagreement structure. We therefore read the 25.5% as evidence of specification sensitivity within the evaluated reasoning model rather than as a strictly like-for-like cross-tier comparison. The arm is documented as the other tiers are — post-hoc label, a dedicated results table plus a new row in the model-snapshots table, a SHA-256 manifest entry, and a data-availability line (DeepSeek-R1 via the OpenRouter API, queried 2026-09-06).

Specific questions

Q1.

given that the 32-specification grid is a d-optimal fraction that is resolution-limited and cannot cleanly separate main effects from two-factor interactions, how do the authors justify interpreting the variance component estimates as reflecting the causal contribution of particular design axes? the authors report, for example, that “answer presentation contributes $\sim 4.6\%$ ” of variance->but this estimate is conditional on the assumed main-effects model and could be substantially biased by aliasing with interactions or by the non-orthogonality of the design. what specific diagnostic tests did the authors conduct to assess the degree of aliasing in their design, and

how would the variance decomposition change if the authors instead used a bayesian hierarchical model with shrinkage on the interaction terms?

(Addressed in manuscript Section 3.1, Table 1, Table 2, and Section 5.2.)

- We do not interpret the components causally; they are descriptive partitions under the assumed model, and we have removed the inferential CIs from the individual per-axis shares.
- Diagnostic conducted: correlation structure of the effect-coded design matrix — main-effect columns near-orthogonal (**0.062**), with the maximum correlation between a main effect and a two-factor interaction of other factors **0.333**, confirming non-negligible aliasing remains; the decomposition is therefore kept descriptive only.
- The paper’s conclusions do not rest on the decomposition: the headline flip, agreement, permutation, and score results are model-free and assume no orthogonality. The decomposition is retained only as a descriptive summary.
- We report the range of variance explained across model specifications (main-effects R^2 0.083; +two-way in-sample 0.119; regularized issuer-out-of-sample 0.101), which shifts with the modelling choice and stays small throughout — a direct illustration of the non-identification, and our answer to the “range of variance explained” request (new table in §3.1 of the revised manuscript).
- On the shrinkage sub-question specifically: we did not fit a Bayesian hierarchical model, because under the descriptive reframing we no longer make an inferential decomposition. A shrinkage fit would sharpen estimates of components we deliberately no longer interpret; reporting the aliasing directly and relying on the model-free quantities is the more honest response to the identification problem the reviewer raises.

Q2.

the real-firm decontamination procedure multiplies all dollar-denominated quantities by an issuer-specific factor sampled from log-uniform[0.6, 1.7], but this transformation changes the scale of all monetary variables and may alter the credit-risk signal in ways that are not controlled for. can the authors demonstrate that the transformed profiles preserve the same ordering on credit-risk metrics as the original profiles? specifically, if the authors compute altman z'' on the transformed profiles, do the z'' values correlate with the original z'' values, and do the transformed profiles remain in the same z'' stratum (safe/watch/distress) with high probability? if the transformation systematically changes the z'' distribution, then the real-firm comparison to the constructed battery is not a comparison of similar credit-risk profiles.

(Addressed in manuscript Section 3.6.)

- Yes. Uniform dollar scaling preserves all ratios, so Z'' is invariant: the correlation between transformed and original Z'' is **1.000000**, and **100%** of issuers retain their original stratum. (Whether the *LLM*’s output responds to absolute scale is a separate question, addressed under Required revision 4.)

Q3.

The specification-instability score (sis) is validated on the same data used to define it, raising concerns about circularity. what evidence can the authors provide that the sis generalizes to specifications not in the original 32-specification grid, such as different prompt paraphrases, different output formats, or different temperature settings? if the authors were to conduct a cross-validation where the sis is computed on a subset of specifications and evaluated on held-out specifications, how stable is the roc-auc across different splits, and how does the choice of the specification set affect the score’s calibration (i.e., does the score depend on the particular specifications included)?

(Addressed in manuscript Section 4.2, Algorithm 1 (Figure 5), and Table 7.)

- Held-out-specification cross-validation ROC-AUC **0.94** (95% CI 0.86–0.99); median **0.88** across 300 balanced splits (IQR 0.86–0.91); external-issuer **0.70** (95% CI 0.51–0.87). Cross-model transfer is moderate but inconsistent across model pairs (per-item Spearman 0.14–0.73, median ≈ 0.45 ; AUC ≈ 0.72) and therefore too variable to rely on, with the recalibration cost stated (re-score the grid on the new model, Table 7). On the threshold: at $k = 16$ the attainable SIS values are discrete, so the Youden-optimal $\tau = \mathbf{0.08}$ reduces to the “flag any disagreement” rule (precision 1.00, recall 0.88 on held-out); we present it as descriptive (chosen and evaluated on the same split) and k -dependent, and direct practitioners to apply the same rule at their deployment k .

Q4.

the pillar-1 capital illustration maps credit ratings through the basel cre20 risk-weight schedule and applies the resulting risk weights to a hypothetical \$45 billion equal-weight portfolio. however, the basel standardized approach requires specification of several portfolio characteristics that are not provided (e.g., seniority, collateral, maturity, country of issuance). how sensitive is the reported capital range (\$2.0–\$4.1 billion) to reasonable variations in these portfolio characteristics? if the authors vary the assumed maturity from 1 to 10 years or assume a 20% instead of 0% collateralization, does the capital range change materially, and does the relative dispersion (interquartile range relative to median) remain at $\sim 18\%$? moreover, given that the basel risk weights are designed for bank portfolios, not for credit-rating evaluation, what justifies the use of this mapping in the context of this study, and how should readers interpret the comparison to the basel rcap exercise in light of these differences?

(Addressed in manuscript Section 3.5, Table 3, and Methods Section 6.3.)

- We appreciate the reviewer highlighting these dependencies, which we agree are difficult to justify. In response, and consistent with our answer to Required revision 6, we have removed the entire Pillar-1 capital illustration, the spread and turnover translations, and the RCAP comparison from the paper; the revised manuscript no longer relies on any Basel risk-weight mapping. We therefore no longer make any claim whose validity depends on the portfolio assumptions the reviewer identifies.